Program Design

Section 3

Chapter 4

An Introduction to Programming Using Python
David I. Schneider

Top-Down Design

- To make a complicated problem more understandable
  - Divide it into smaller, less complex subproblems.
  - Called stepwise refinement
- Top-down design and structured programming
  - Techniques to enhance programming productivity

Structured Programming

- Meets modern standards of program design
- Use top-down design
- Use only the three types of logical structures:
  - **Sequences** – Statements executed one after the other
  - **Decisions** – blocks of code executed based on test of some condition
  - **Loops** – blocks of coded executed repeatedly based on some condition

Advantages of Structured Programming

- Easy to write
- Easy to debug
- Easy to understand
- Easy to change

Object-Oriented Programming

- An object is an encapsulation of data and code that operates on the data
- Objects
  - Have properties
  - Respond to methods
  - Raise events
- Object-oriented program viewed as collection of cooperating objects

Object-Oriented Programming

- Python is an object-oriented programming language;
  - Every structure such as a list or a string is actually an object.
  - We have been learning the building blocks of Python, later we'll put them together using object-oriented techniques in Chapter 7.

A Relevant Quote

From *Dirk Gently's Holistic Detective Agency*

"… the more slow and dim-witted your pupil, the more you have to break things down into more and more simple ideas. And that's really the essence of programming."

End

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